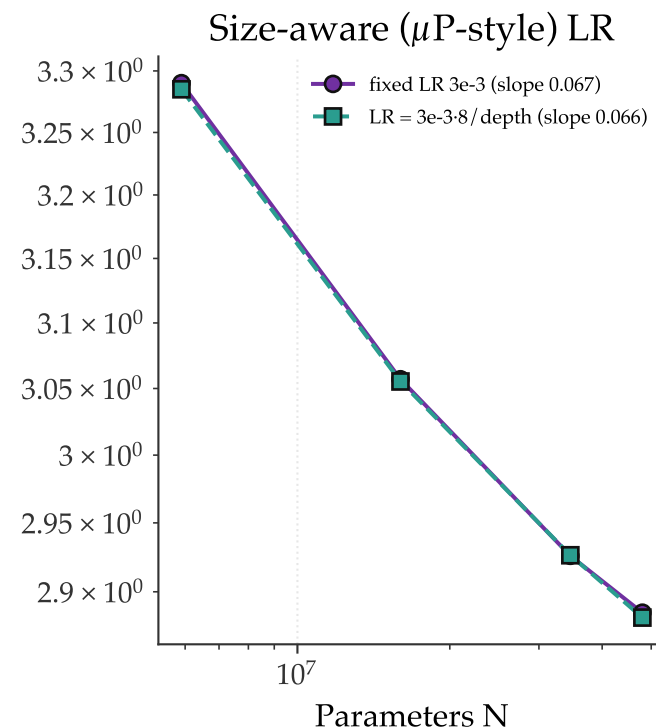
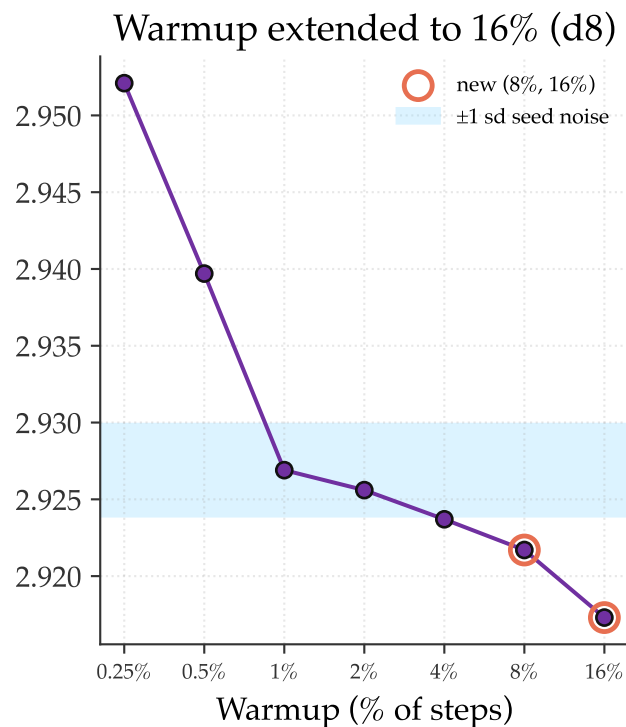
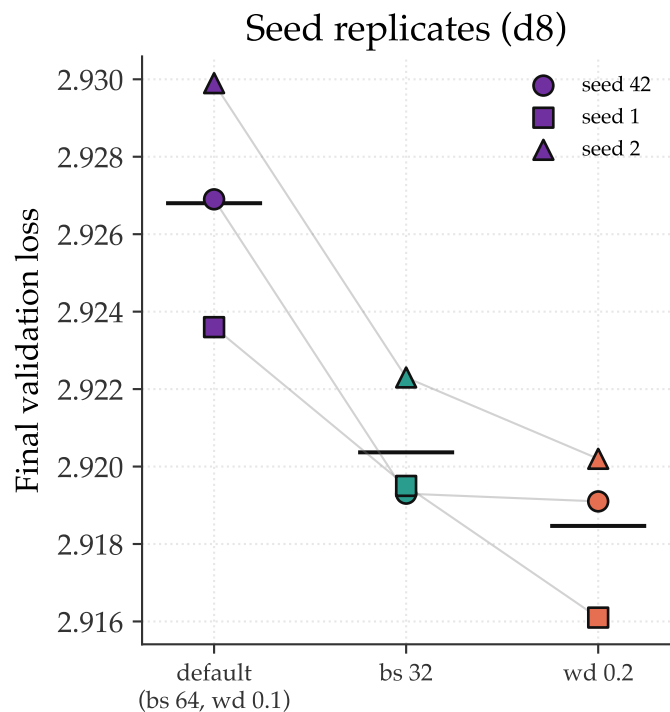


Follow-ups: seed replicates, warmup bracket, size-aware LR (9 runs)



Seed replicates: bs 32 and wd 0.2 beat the default at every seed. Mean gains -0.006 (paired $t = 5.5$) and -0.008 ($t = 12$).

Both small 1(a) wins are real, not noise. Seed noise of the default itself: sd 0.003 (range 0.006).

Warmup: 8% $\rightarrow$ 2.922, 16% $\rightarrow$ 2.917. Still slowly improving, so there is no sharp optimum; beyond ~1% warmup is a flat, safe plateau (0.01 total gain from 1% to 16%, each step near noise). Warmup matters far more at high LR (see 1(b)).

Size-aware LR (LR = $3e-3 \times 8 / \text{depth}$, μ P-style): within 0.005 of fixed LR at every size; slope 0.066 vs 0.067.

The default LR is already near-optimal across d4-d9, so the 2(a) baseline slope is not an LR-mistuning artifact.

Size-aware LR matters when LR is far from optimal (LR 0.03's gap grew 0.03 $\rightarrow$ 0.11 with size), not when it is close.